

WASP-6b

R-2140

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-6_250903 · created 2026-08-12T00:14:52+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460921.8497 +/- 0.0038 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.193 +/- 0.029
Transit depth (Rp/Rs)^2	3.72 +/- 1.11 [%]
Orbital Inclination (inc)	87.78 +/- 1.29
Airmass coefficient 1 (a1)	1.447 +/- 0.088
Airmass coefficient 2 (a2)	-0.195 +/- 0.088
Scatter in the residuals of the lightcurve fit is	6.11 %
Best Comparison Star	None
Optimal Aperture	3.0
Optimal Annulus	7.5
Transit Duration (day)	0.106 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.193 ± 0.029 vs 0.1446 (deviation 1.29σ)
Duration fit vs published	0.106 d vs 0.1087 d (deviation 0.19σ)
Mid-time O–C	15.1 min
Depth SNR	5.2
Residual scatter	6.11 %

Light curve

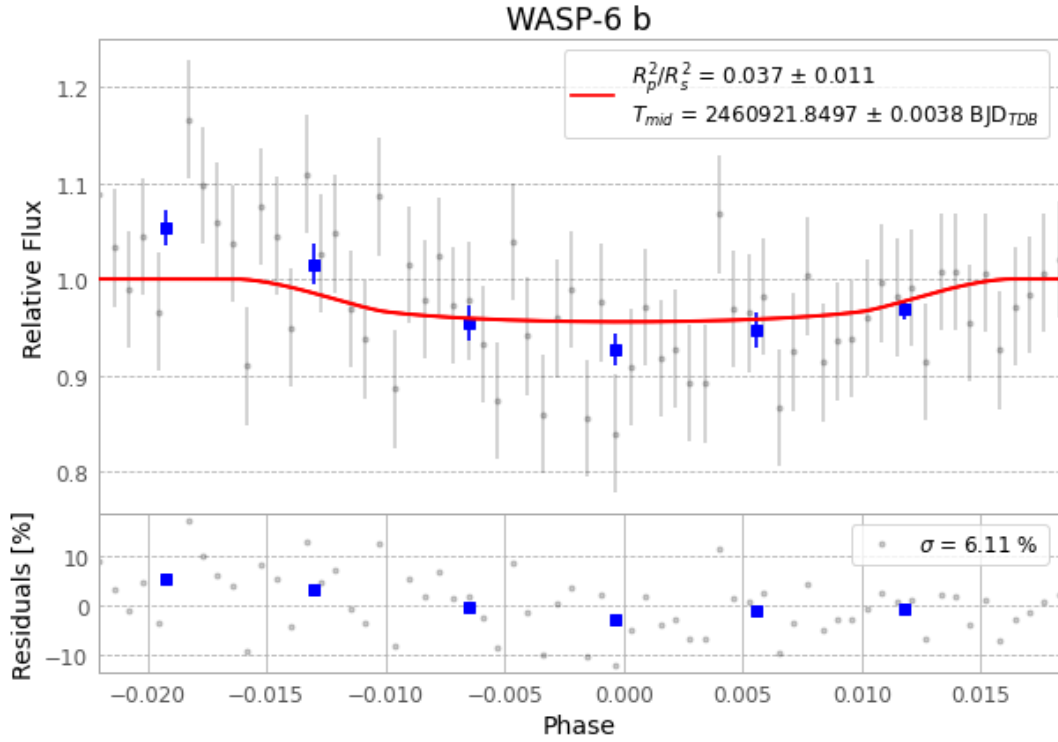


Figure 1 — FinalLightCurve_WASP-6 b_2025-09-02.png (SHA-256 9cacf0eaf9bc6a3a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [577, 346], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0d6b4b614b2c5709...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

66 FITS frames (43 MB), WASP-6250903062715.FITS → WASP-6250903094215.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-6-250903.log (2fa982120f5f...), FinalLightCurve_WASP-6 b_2025-09-02.png (9cacf0eaf9bc...), FinalLightCurve_WASP-6 b_2025-09-02.csv (d69c91157c8b...)

Compute resources

Wall time 257.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-12 00:14Z.